

September 23, 2022

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Fall Biosolids
Pace Project No.: 50325469

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on September 09, 2022. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Fall Biosolids

Pace Project No.: 50325469

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268

Illinois Accreditation #: 200074

Indiana Drinking Water Laboratory #: C-49-06

Kansas/TNI Certification #: E-10177

Kentucky UST Agency Interest #: 80226

Kentucky WW Laboratory ID #: 98019

Michigan Drinking Water Laboratory #9050

Ohio VAP Certified Laboratory #: CL0065

Oklahoma Laboratory #: 9204

Texas Certification #: T104704355

Wisconsin Laboratory #: 999788130

USDA Soil Permit #: P330-19-00257

REPORT OF LABORATORY ANALYSIS

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SAMPLE SUMMARY

Project: Fall Biosolids

Pace Project No.: 50325469

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50325469001	Digestor #4	Solid	09/08/22 13:25	09/09/22 11:30

REPORT OF LABORATORY ANALYSIS

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SAMPLE ANALYTE COUNT

Project: Fall Biosolids

Pace Project No.: 50325469

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50325469001	Digestor #4	EPA 8081	KAV	21	PASI-I
		EPA 8082	RID	8	PASI-I
		EPA 8151A	BJW	3	PASI-I
		EPA 6010	MTM	6	PASI-I
		EPA 6010	JPk	8	PASI-I
		EPA 6020	CAW	10	PASI-I
		EPA 7470	ILP	1	PASI-I
		EPA 7471	ILP	1	PASI-I
		EPA 8270	JCM	67	PASI-I
		EPA 8270	JCM	18	PASI-I
		EPA 8260	TLS1	55	PASI-I
		EPA 5030/8260	BES	13	PASI-I
		SM 2540G	MMS	1	PASI-I
		SM 2540G	MMS	1	PASI-I
		SM 2540G	MMS	1	PASI-I
		EPA 9038	BEP	1	PASI-I
		EPA 9045	TRK	1	PASI-I
		EPA 351.2	STS	1	PASI-I
		EPA 353.2	OAS	2	PASI-I
		SM 4500-CI-E	OAS	1	PASI-I
		SM 4500-CN-E	ZM	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Fall Biosolids
Pace Project No.: 50325469

Sample: Digestor #4 **Lab ID: 50325469001** Collected: 09/08/22 13:25 Received: 09/09/22 11:30 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8081 GCS Pesticide Solids								
Analytical Method: EPA 8081 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Aldrin	<1480	ug/kg	1480	1	09/14/22 09:40	09/14/22 22:50	309-00-2	
alpha-BHC	<1480	ug/kg	1480	1	09/14/22 09:40	09/14/22 22:50	319-84-6	
beta-BHC	<1480	ug/kg	1480	1	09/14/22 09:40	09/14/22 22:50	319-85-7	
delta-BHC	<1480	ug/kg	1480	1	09/14/22 09:40	09/14/22 22:50	319-86-8	
gamma-BHC (Lindane)	<1480	ug/kg	1480	1	09/14/22 09:40	09/14/22 22:50	58-89-9	
Chlordane (Technical)	<29600	ug/kg	29600	1	09/14/22 09:40	09/14/22 22:50	57-74-9	
alpha-Chlordane	<1480	ug/kg	1480	1	09/14/22 09:40	09/14/22 22:50	5103-71-9	
gamma-Chlordane	<1480	ug/kg	1480	1	09/14/22 09:40	09/14/22 22:50	5103-74-2	
4,4'-DDD	<2960	ug/kg	2960	1	09/14/22 09:40	09/14/22 22:50	72-54-8	
4,4'-DDE	<2960	ug/kg	2960	1	09/14/22 09:40	09/14/22 22:50	72-55-9	
4,4'-DDT	<2960	ug/kg	2960	1	09/14/22 09:40	09/14/22 22:50	50-29-3	
Dieldrin	<2960	ug/kg	2960	1	09/14/22 09:40	09/14/22 22:50	60-57-1	
Endosulfan I	<1480	ug/kg	1480	1	09/14/22 09:40	09/14/22 22:50	959-98-8	
Endosulfan II	<2960	ug/kg	2960	1	09/14/22 09:40	09/14/22 22:50	33213-65-9	
Endosulfan sulfate	<2960	ug/kg	2960	1	09/14/22 09:40	09/14/22 22:50	1031-07-8	
Endrin	<2960	ug/kg	2960	1	09/14/22 09:40	09/14/22 22:50	72-20-8	
Endrin aldehyde	<2960	ug/kg	2960	1	09/14/22 09:40	09/14/22 22:50	7421-93-4	
Heptachlor	<1480	ug/kg	1480	1	09/14/22 09:40	09/14/22 22:50	76-44-8	
Heptachlor epoxide	<1480	ug/kg	1480	1	09/14/22 09:40	09/14/22 22:50	1024-57-3	
Toxaphene	<29600	ug/kg	29600	1	09/14/22 09:40	09/14/22 22:50	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	38	%.	10-147	1	09/14/22 09:40	09/14/22 22:50	2051-24-3	
8082 PCB Solids								
Analytical Method: EPA 8082 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
PCB-1016 (Aroclor 1016)	<37400	ug/kg	37400	1	09/14/22 10:26	09/15/22 13:30	12674-11-2	
PCB-1221 (Aroclor 1221)	<37400	ug/kg	37400	1	09/14/22 10:26	09/15/22 13:30	11104-28-2	
PCB-1232 (Aroclor 1232)	<37400	ug/kg	37400	1	09/14/22 10:26	09/15/22 13:30	11141-16-5	
PCB-1242 (Aroclor 1242)	<37400	ug/kg	37400	1	09/14/22 10:26	09/15/22 13:30	53469-21-9	
PCB-1248 (Aroclor 1248)	<37400	ug/kg	37400	1	09/14/22 10:26	09/15/22 13:30	12672-29-6	
PCB-1254 (Aroclor 1254)	<37400	ug/kg	37400	1	09/14/22 10:26	09/15/22 13:30	11097-69-1	
PCB-1260 (Aroclor 1260)	<37400	ug/kg	37400	1	09/14/22 10:26	09/15/22 13:30	11096-82-5	
Surrogates								
Tetrachloro-m-xylene (S)	16	%.	36-112	1	09/14/22 10:26	09/15/22 13:30	877-09-8	S2
8151A CI Acid Herbicides TCLP								
Analytical Method: EPA 8151A Preparation Method: EPA 8151								
Leachate Method/Date: EPA 1311; 09/16/22 15:00 Initial pH: 8.02; Final pH: 5.17								
Pace Analytical Services - Indianapolis								
2,4-D	<0.0050	mg/L	0.0050	1	09/20/22 09:36	09/21/22 17:45	94-75-7	
2,4,5-TP (Silvex)	<0.0050	mg/L	0.0050	1	09/20/22 09:36	09/21/22 17:45	93-72-1	
Surrogates								
2,4-DCAA (S)	86	%.	12-143	1	09/20/22 09:36	09/21/22 17:45	19719-28-9	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Fall Biosolids
Pace Project No.: 50325469

Sample: Digestor #4 **Lab ID: 50325469001** Collected: 09/08/22 13:25 Received: 09/09/22 11:30 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
6010 MET ICP								
Analytical Method: EPA 6010 Preparation Method: EPA 3050								
Pace Analytical Services - Indianapolis								
Aluminum	<4720	mg/kg	4720	1	09/15/22 14:35	09/16/22 15:49	7429-90-5	
Calcium	25200	mg/kg	4720	1	09/15/22 14:35	09/16/22 15:49	7440-70-2	
Magnesium	4890	mg/kg	4720	1	09/15/22 14:35	09/16/22 15:49	7439-95-4	
Phosphorus	21100	mg/kg	4720	1	09/15/22 14:35	09/16/22 15:49	7723-14-0	N2
Potassium	5660	mg/kg	4720	1	09/15/22 14:35	09/16/22 15:49	7440-09-7	
Sodium	8750	mg/kg	4720	1	09/15/22 14:35	09/16/22 15:49	7440-23-5	
6010 MET ICP, TCLP								
Analytical Method: EPA 6010 Preparation Method: EPA 3010								
Leachate Method/Date: EPA 1311; 09/16/22 15:00 Initial pH: 8.02; Final pH: 5.17								
Pace Analytical Services - Indianapolis								
Arsenic	<0.10	mg/L	0.10	1	09/19/22 13:25	09/20/22 02:05	7440-38-2	
Barium	<5.0	mg/L	5.0	1	09/19/22 13:25	09/20/22 02:05	7440-39-3	
Cadmium	<0.050	mg/L	0.050	1	09/19/22 13:25	09/20/22 02:05	7440-43-9	
Chromium	<0.10	mg/L	0.10	1	09/19/22 13:25	09/20/22 02:05	7440-47-3	
Lead	<0.10	mg/L	0.10	1	09/19/22 13:25	09/20/22 02:05	7439-92-1	
Selenium	<0.10	mg/L	0.10	1	09/19/22 13:25	09/20/22 02:05	7782-49-2	
Silver	<0.10	mg/L	0.10	1	09/19/22 13:25	09/20/22 02:05	7440-22-4	
Zinc	2.3	mg/L	0.20	1	09/19/22 13:25	09/20/22 02:05	7440-66-6	
6020 MET ICPMS								
Analytical Method: EPA 6020 Preparation Method: EPA 3050B								
Pace Analytical Services - Indianapolis								
Arsenic	31.4	mg/kg	8.9	1	09/16/22 09:15	09/19/22 16:13	7440-38-2	
Cadmium	10.5	mg/kg	4.4	1	09/16/22 09:15	09/19/22 16:13	7440-43-9	
Chromium	208	mg/kg	17.8	1	09/16/22 09:15	09/19/22 16:13	7440-47-3	
Copper	1490	mg/kg	8.9	1	09/16/22 09:15	09/19/22 16:13	7440-50-8	
Lead	75.6	mg/kg	8.9	1	09/16/22 09:15	09/19/22 16:13	7439-92-1	
Molybdenum	34.1	mg/kg	8.9	1	09/16/22 09:15	09/19/22 16:13	7439-98-7	N2
Nickel	46.3	mg/kg	4.4	1	09/16/22 09:15	09/19/22 16:13	7440-02-0	
Selenium	10	mg/kg	8.9	1	09/16/22 09:15	09/19/22 16:13	7782-49-2	
Silver	8.0	mg/kg	4.4	1	09/16/22 09:15	09/19/22 16:13	7440-22-4	
Zinc	4010	mg/kg	222	5	09/16/22 09:15	09/19/22 16:09	7440-66-6	
7470 Mercury, TCLP								
Analytical Method: EPA 7470 Preparation Method: EPA 7470								
Leachate Method/Date: EPA 1311; 09/16/22 15:00 Initial pH: 8.02; Final pH: 5.17								
Pace Analytical Services - Indianapolis								
Mercury	<0.0020	mg/L	0.0020	1	09/20/22 16:56	09/21/22 08:16	7439-97-6	
7471 Mercury								
Analytical Method: EPA 7471 Preparation Method: EPA 7471								
Pace Analytical Services - Indianapolis								
Mercury	<17.9	mg/kg	17.9	1	09/19/22 19:06	09/20/22 10:45	7439-97-6	
8270 SVOC SS Soil								
Analytical Method: EPA 8270 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Acenaphthene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	83-32-9	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Fall Biosolids

Pace Project No.: 50325469

Sample: Digestor #4 Lab ID: 50325469001 Collected: 09/08/22 13:25 Received: 09/09/22 11:30 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 SVOC SS Soil Analytical Method: EPA 8270 Preparation Method: EPA 3546 Pace Analytical Services - Indianapolis								
Acenaphthylene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	208-96-8	
Anthracene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	120-12-7	
Benzo(a)anthracene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	56-55-3	
Benzo(a)pyrene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	50-32-8	
Benzo(b)fluoranthene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	205-99-2	
Benzo(g,h,i)perylene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	191-24-2	
Benzo(k)fluoranthene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	207-08-9	
Benzyl alcohol	<937000	ug/kg	937000	1	09/20/22 09:40	09/20/22 16:43	100-51-6	
4-Bromophenylphenyl ether	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	101-55-3	
Butylbenzylphthalate	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	85-68-7	
4-Chloro-3-methylphenol	<937000	ug/kg	937000	1	09/20/22 09:40	09/20/22 16:43	59-50-7	
4-Chloroaniline	<937000	ug/kg	937000	1	09/20/22 09:40	09/20/22 16:43	106-47-8	
bis(2-Chloroethoxy)methane	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	111-91-1	
bis(2-Chloroethyl) ether	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	111-44-4	
bis(2chloro1methylethyl) ether	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	108-60-1	
2-Chloronaphthalene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	91-58-7	
2-Chlorophenol	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	95-57-8	
4-Chlorophenylphenyl ether	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	7005-72-3	
Chrysene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	218-01-9	
Dibenz(a,h)anthracene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	53-70-3	
Dibenzofuran	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	132-64-9	
3,3'-Dichlorobenzidine	<937000	ug/kg	937000	1	09/20/22 09:40	09/20/22 16:43	91-94-1	
2,4-Dichlorophenol	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	120-83-2	
Diethylphthalate	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	84-66-2	
2,4-Dimethylphenol	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	105-67-9	
Dimethylphthalate	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	131-11-3	
Di-n-butylphthalate	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	84-74-2	
4,6-Dinitro-2-methylphenol	<937000	ug/kg	937000	1	09/20/22 09:40	09/20/22 16:43	534-52-1	
2,4-Dinitrophenol	<2270000	ug/kg	2270000	1	09/20/22 09:40	09/20/22 16:43	51-28-5	
2,4-Dinitrotoluene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	121-14-2	
2,6-Dinitrotoluene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	606-20-2	
Di-n-octylphthalate	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	117-84-0	
bis(2-Ethylhexyl)phthalate	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	117-81-7	
Fluoranthene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	206-44-0	
Fluorene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	86-73-7	
Hexachloro-1,3-butadiene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	87-68-3	
Hexachlorobenzene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	118-74-1	
Hexachlorocyclopentadiene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	77-47-4	
Hexachloroethane	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	67-72-1	
Indeno(1,2,3-cd)pyrene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	193-39-5	
Isophorone	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	78-59-1	
1-Methylnaphthalene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	90-12-0	
2-Methylnaphthalene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	91-57-6	
2-Methylphenol(o-Cresol)	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	95-48-7	
3&4-Methylphenol(m&p Cresol)	<937000	ug/kg	937000	1	09/20/22 09:40	09/20/22 16:43		

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Fall Biosolids
Pace Project No.: 50325469

Sample: Digestor #4 **Lab ID: 50325469001** Collected: 09/08/22 13:25 Received: 09/09/22 11:30 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 SVOC SS Soil								
Analytical Method: EPA 8270 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Naphthalene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	91-20-3	
2-Nitroaniline	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	88-74-4	
3-Nitroaniline	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	99-09-2	
4-Nitroaniline	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	100-01-6	
Nitrobenzene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	98-95-3	
2-Nitrophenol	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	88-75-5	
4-Nitrophenol	<2270000	ug/kg	2270000	1	09/20/22 09:40	09/20/22 16:43	100-02-7	
N-Nitroso-di-n-propylamine	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	621-64-7	
N-Nitrosodiphenylamine	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	86-30-6	
Pentachlorophenol	<2270000	ug/kg	2270000	1	09/20/22 09:40	09/20/22 16:43	87-86-5	
Phenanthrene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	85-01-8	
Phenol	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	108-95-2	P1
Pyrene	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	129-00-0	
2,4,5-Trichlorophenol	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	95-95-4	
2,4,6-Trichlorophenol	<469000	ug/kg	469000	1	09/20/22 09:40	09/20/22 16:43	88-06-2	

Surrogates

Nitrobenzene-d5 (S)	76	%.	35-110	1	09/20/22 09:40	09/20/22 16:43	4165-60-0	
Phenol-d5 (S)	102	%.	35-115	1	09/20/22 09:40	09/20/22 16:43	4165-62-2	
2-Fluorophenol (S)	95	%.	22-114	1	09/20/22 09:40	09/20/22 16:43	367-12-4	
2,4,6-Tribromophenol (S)	97	%.	10-123	1	09/20/22 09:40	09/20/22 16:43	118-79-6	
2-Fluorobiphenyl (S)	77	%.	36-100	1	09/20/22 09:40	09/20/22 16:43	321-60-8	
p-Terphenyl-d14 (S)	88	%.	29-117	1	09/20/22 09:40	09/20/22 16:43	1718-51-0	

8270 MSSV TCLP Sep Funnel

Analytical Method: EPA 8270 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 09/16/22 15:00 Initial pH: 8.02; Final pH: 5.17

Pace Analytical Services - Indianapolis

1,4-Dichlorobenzene	<0.29	mg/L	0.29	1	09/20/22 13:00	09/21/22 23:08	106-46-7	
2,4-Dinitrotoluene	<0.29	mg/L	0.29	1	09/20/22 13:00	09/21/22 23:08	121-14-2	
Hexachloro-1,3-butadiene	<0.29	mg/L	0.29	1	09/20/22 13:00	09/21/22 23:08	87-68-3	
Hexachlorobenzene	<0.29	mg/L	0.29	1	09/20/22 13:00	09/21/22 23:08	118-74-1	
Hexachloroethane	<0.29	mg/L	0.29	1	09/20/22 13:00	09/21/22 23:08	67-72-1	
2-Methylphenol(o-Cresol)	<0.29	mg/L	0.29	1	09/20/22 13:00	09/21/22 23:08	95-48-7	
3&4-Methylphenol(m&p Cresol)	<0.57	mg/L	0.57	1	09/20/22 13:00	09/21/22 23:08		
Nitrobenzene	<0.29	mg/L	0.29	1	09/20/22 13:00	09/21/22 23:08	98-95-3	
Pentachlorophenol	<1.4	mg/L	1.4	1	09/20/22 13:00	09/21/22 23:08	87-86-5	
Pyridine	<0.29	mg/L	0.29	1	09/20/22 13:00	09/21/22 23:08	110-86-1	
2,4,5-Trichlorophenol	<1.4	mg/L	1.4	1	09/20/22 13:00	09/21/22 23:08	95-95-4	
2,4,6-Trichlorophenol	<0.29	mg/L	0.29	1	09/20/22 13:00	09/21/22 23:08	88-06-2	

Surrogates

Nitrobenzene-d5 (S)	69	%.	51-109	1	09/20/22 13:00	09/21/22 23:08	4165-60-0	
2-Fluorobiphenyl (S)	62	%.	38-101	1	09/20/22 13:00	09/21/22 23:08	321-60-8	
p-Terphenyl-d14 (S)	88	%.	65-159	1	09/20/22 13:00	09/21/22 23:08	1718-51-0	
Phenol-d5 (S)	28	%.	16-44	1	09/20/22 13:00	09/21/22 23:08	4165-62-2	
2-Fluorophenol (S)	44	%.	24-70	1	09/20/22 13:00	09/21/22 23:08	367-12-4	
2,4,6-Tribromophenol (S)	96	%.	48-137	1	09/20/22 13:00	09/21/22 23:08	118-79-6	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Fall Biosolids

Pace Project No.: 50325469

Sample: Digestor #4 **Lab ID: 50325469001** Collected: 09/08/22 13:25 Received: 09/09/22 11:30 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV 5030 Low Level		Analytical Method: EPA 8260 Pace Analytical Services - Indianapolis						
Acetone	<9.4	mg/kg	9.4	1		09/19/22 12:05	67-64-1	
Acrolein	<9.4	mg/kg	9.4	1		09/19/22 12:05	107-02-8	
Acrylonitrile	<9.4	mg/kg	9.4	1		09/19/22 12:05	107-13-1	
Benzene	<0.47	mg/kg	0.47	1		09/19/22 12:05	71-43-2	
Bromodichloromethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	75-27-4	
Bromoform	<0.47	mg/kg	0.47	1		09/19/22 12:05	75-25-2	
Bromomethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	74-83-9	
2-Butanone (MEK)	2.9	mg/kg	2.4	1		09/19/22 12:05	78-93-3	CH,H7
Carbon disulfide	<0.94	mg/kg	0.94	1		09/19/22 12:05	75-15-0	
Carbon tetrachloride	<0.47	mg/kg	0.47	1		09/19/22 12:05	56-23-5	
Chlorobenzene	<0.47	mg/kg	0.47	1		09/19/22 12:05	108-90-7	
Chloroethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	75-00-3	
Chloroform	<0.47	mg/kg	0.47	1		09/19/22 12:05	67-66-3	
Chloromethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	74-87-3	
1,2-Dibromo-3-chloropropane	<0.94	mg/kg	0.94	1		09/19/22 12:05	96-12-8	
Dibromochloromethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	124-48-1	
1,2-Dibromoethane (EDB)	<0.47	mg/kg	0.47	1		09/19/22 12:05	106-93-4	
Dibromomethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	74-95-3	
1,2-Dichlorobenzene	<0.47	mg/kg	0.47	1		09/19/22 12:05	95-50-1	
1,3-Dichlorobenzene	<0.47	mg/kg	0.47	1		09/19/22 12:05	541-73-1	
1,4-Dichlorobenzene	<0.47	mg/kg	0.47	1		09/19/22 12:05	106-46-7	
trans-1,4-Dichloro-2-butene	<9.4	mg/kg	9.4	1		09/19/22 12:05	110-57-6	
Dichlorodifluoromethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	75-71-8	L1
1,1-Dichloroethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	75-34-3	
1,2-Dichloroethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	107-06-2	
1,1-Dichloroethene	<0.47	mg/kg	0.47	1		09/19/22 12:05	75-35-4	
cis-1,2-Dichloroethene	<0.47	mg/kg	0.47	1		09/19/22 12:05	156-59-2	
trans-1,2-Dichloroethene	<0.47	mg/kg	0.47	1		09/19/22 12:05	156-60-5	
1,2-Dichloropropane	<0.47	mg/kg	0.47	1		09/19/22 12:05	78-87-5	
cis-1,3-Dichloropropene	<0.47	mg/kg	0.47	1		09/19/22 12:05	10061-01-5	
trans-1,3-Dichloropropene	<0.47	mg/kg	0.47	1		09/19/22 12:05	10061-02-6	
Ethylbenzene	<0.47	mg/kg	0.47	1		09/19/22 12:05	100-41-4	
Ethyl methacrylate	<9.4	mg/kg	9.4	1		09/19/22 12:05	97-63-2	
2-Hexanone	<9.4	mg/kg	9.4	1		09/19/22 12:05	591-78-6	
Iodomethane	<9.4	mg/kg	9.4	1		09/19/22 12:05	74-88-4	
Methylene Chloride	<1.9	mg/kg	1.9	1		09/19/22 12:05	75-09-2	
Methyl methacrylate	<9.4	mg/kg	9.4	1		09/19/22 12:05	80-62-6	
4-Methyl-2-pentanone (MIBK)	<2.4	mg/kg	2.4	1		09/19/22 12:05	108-10-1	
Methyl-tert-butyl ether	<0.47	mg/kg	0.47	1		09/19/22 12:05	1634-04-4	
Styrene	<0.47	mg/kg	0.47	1		09/19/22 12:05	100-42-5	
1,1,1,2-Tetrachloroethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	630-20-6	
1,1,2,2-Tetrachloroethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	79-34-5	
Tetrachloroethene	<0.47	mg/kg	0.47	1		09/19/22 12:05	127-18-4	
Toluene	<0.47	mg/kg	0.47	1		09/19/22 12:05	108-88-3	
1,1,1-Trichloroethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	71-55-6	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Fall Biosolids
Pace Project No.: 50325469

Sample: Digestor #4 **Lab ID: 50325469001** Collected: 09/08/22 13:25 Received: 09/09/22 11:30 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV 5030 Low Level								
Analytical Method: EPA 8260								
Pace Analytical Services - Indianapolis								
1,1,2-Trichloroethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	79-00-5	
Trichloroethene	<0.47	mg/kg	0.47	1		09/19/22 12:05	79-01-6	
Trichlorofluoromethane	<0.47	mg/kg	0.47	1		09/19/22 12:05	75-69-4	L1
1,2,3-Trichloropropane	<0.47	mg/kg	0.47	1		09/19/22 12:05	96-18-4	
Vinyl acetate	<9.4	mg/kg	9.4	1		09/19/22 12:05	108-05-4	L1
Vinyl chloride	<0.47	mg/kg	0.47	1		09/19/22 12:05	75-01-4	
Xylene (Total)	<0.94	mg/kg	0.94	1		09/19/22 12:05	1330-20-7	
Surrogates								
Dibromofluoromethane (S)	106	%	62-146	1		09/19/22 12:05	1868-53-7	
Toluene-d8 (S)	100	%	68-143	1		09/19/22 12:05	2037-26-5	
4-Bromofluorobenzene (S)	84	%	63-129	1		09/19/22 12:05	460-00-4	
8260 MSV TCLP								
Analytical Method: EPA 5030/8260 Leachate Method/Date: EPA 1311; 09/16/22 14:40								
Pace Analytical Services - Indianapolis								
Benzene	<0.050	mg/L	0.050	1		09/19/22 15:12	71-43-2	
2-Butanone (MEK)	<1.0	mg/L	1.0	1		09/19/22 15:12	78-93-3	
Carbon tetrachloride	<0.050	mg/L	0.050	1		09/19/22 15:12	56-23-5	
Chlorobenzene	<0.050	mg/L	0.050	1		09/19/22 15:12	108-90-7	
Chloroform	<0.050	mg/L	0.050	1		09/19/22 15:12	67-66-3	
1,2-Dichloroethane	<0.050	mg/L	0.050	1		09/19/22 15:12	107-06-2	
1,1-Dichloroethene	<0.050	mg/L	0.050	1		09/19/22 15:12	75-35-4	
Tetrachloroethene	<0.050	mg/L	0.050	1		09/19/22 15:12	127-18-4	
Trichloroethene	<0.050	mg/L	0.050	1		09/19/22 15:12	79-01-6	
Vinyl chloride	<0.020	mg/L	0.020	1		09/19/22 15:12	75-01-4	
Surrogates								
4-Bromofluorobenzene (S)	90	%	79-124	1		09/19/22 15:12	460-00-4	
Dibromofluoromethane (S)	112	%	82-128	1		09/19/22 15:12	1868-53-7	
Toluene-d8 (S)	101	%	73-122	1		09/19/22 15:12	2037-26-5	
Percent Moisture								
Analytical Method: SM 2540G								
Pace Analytical Services - Indianapolis								
Percent Moisture	98.9	%	0.10	1		09/14/22 21:47		N2
2540G Total Percent Solids								
Analytical Method: SM 2540G								
Pace Analytical Services - Indianapolis								
Total Solids	1.1	%	0.10	1		09/14/22 21:47		N2
2540G Total Volatile Solids								
Analytical Method: SM 2540G								
Pace Analytical Services - Indianapolis								
Total Volatile Solids	54.7	%	0.10	1		09/14/22 21:52		N2
9038 Sulfate Soil								
Analytical Method: EPA 9038 Preparation Method: EPA 9038								
Pace Analytical Services - Indianapolis								
Sulfate	<9460	mg/kg	9460	1	09/13/22 09:10	09/13/22 13:07	14808-79-8	N2

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Fall Biosolids
Pace Project No.: 50325469

Sample: Digester #4 **Lab ID: 50325469001** Collected: 09/08/22 13:25 Received: 09/09/22 11:30 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
9045 pH Soil								
Analytical Method: EPA 9045 Pace Analytical Services - Indianapolis								
pH at 25 Degrees C	8.4	Std. Units	0.10	1		09/14/22 17:20		H3
351.2 Total Kjeldahl Nitrogen								
Analytical Method: EPA 351.2 Preparation Method: EPA 351.2 Pace Analytical Services - Indianapolis								
Nitrogen, Kjeldahl, Total	159000	mg/kg	4690	1	09/12/22 09:50	09/13/22 12:31	7727-37-9	N2
353.2 Nitrogen, NO2/NO3								
Analytical Method: EPA 353.2 Preparation Method: EPA 353.2 Pace Analytical Services - Indianapolis								
Nitrogen, NO2 plus NO3	<23200	mg/kg	23200	50	09/14/22 06:57	09/14/22 10:12		D3,N2
Nitrogen, Nitrate	<23200	mg/kg	23200	50	09/14/22 06:57	09/14/22 10:12	14797-55-8	D3,N2
4500 Chloride in Soil								
Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E Pace Analytical Services - Indianapolis								
Chloride	<186000	mg/kg	186000	20	09/14/22 06:57	09/16/22 10:53	16887-00-6	D3,N2
4500CNE Cyanide, Soil								
Analytical Method: SM 4500-CN-E Preparation Method: SM 4500-CN-C Pace Analytical Services - Indianapolis								
Cyanide	<45.3	mg/kg	45.3	1	09/13/22 09:55	09/13/22 13:51	57-12-5	N2

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

QC Batch:	696493	Analysis Method:	EPA 7470
QC Batch Method:	EPA 7470	Analysis Description:	7470 Mercury TCLP
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3202611 Matrix: Water
Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/L	<0.00067	0.00067	09/21/22 08:01	

LABORATORY CONTROL SAMPLE: 3202612

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	0.005	0.0053	106	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3202613 3202614

Parameter	Units	50325435002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/L	ND	0.015	0.015	0.016	0.015	103	102	75-125	1	20	

MATRIX SPIKE SAMPLE: 3202615

Parameter	Units	50325469001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	<0.0020	0.015	0.015	98	75-125	

MATRIX SPIKE SAMPLE: 3202616

Parameter	Units	50325597001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.011	76	75-125	

MATRIX SPIKE SAMPLE: 3202617

Parameter	Units	50325598001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	101	75-125	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

MATRIX SPIKE SAMPLE:		3202618					
		50325599001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.016	105	75-125	

MATRIX SPIKE SAMPLE:		3202619					
		50325601001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.016	104	75-125	

MATRIX SPIKE SAMPLE:		3202620					
		50325605001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	100	75-125	

MATRIX SPIKE SAMPLE:		3202621					
		50325975001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Mercury	mg/L	ND	0.005	0.0052	105	75-125	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

QC Batch: 696428

QC Batch Method: EPA 7471

Analysis Method: EPA 7471

Analysis Description: 7471 Mercury

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3202414

Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.19	0.19	09/20/22 09:48	

LABORATORY CONTROL SAMPLE: 3202415

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.49	0.50	102	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3202416 3202417

Parameter	Units	50325527001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	2.0	0.69	0.68	3.1	2.4	158	58	75-125	25	20	M0, R1

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

QC Batch:	695590	Analysis Method:	EPA 6010
QC Batch Method:	EPA 3050	Analysis Description:	6010 MET
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3198186 Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<50.0	50.0	09/16/22 14:57	
Calcium	mg/kg	<50.0	50.0	09/16/22 14:57	
Magnesium	mg/kg	<50.0	50.0	09/16/22 14:57	
Phosphorus	mg/kg	ND	50.0	09/16/22 14:57	N2
Potassium	mg/kg	<50.0	50.0	09/16/22 14:57	
Sodium	mg/kg	<50.0	50.0	09/16/22 14:57	

LABORATORY CONTROL SAMPLE: 3198187

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	500	495	99	80-120	
Calcium	mg/kg	500	512	102	80-120	
Magnesium	mg/kg	500	478	96	80-120	
Phosphorus	mg/kg	50	51.3	103	80-120	N2
Potassium	mg/kg	500	476	95	80-120	
Sodium	mg/kg	500	492	98	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3198188 3198189

Parameter	Units	50325381001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	3350	511	503	5300	5250	383	379	75-125	1	20	P6
Calcium	mg/kg	636	511	503	1660	1470	201	167	75-125	12	20	M3
Magnesium	mg/kg	529	511	503	1310	1170	153	128	75-125	11	20	M3
Phosphorus	mg/kg	184	51.1	50.3	236	265	102	161	75-125	11	20	M0, N2
Potassium	mg/kg	299	511	503	922	920	122	124	75-125	0	20	
Sodium	mg/kg	122	511	503	597	599	93	95	75-125	0	20	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

QC Batch: 696431	Analysis Method: EPA 6010
QC Batch Method: EPA 3010	Analysis Description: 6010 MET TCLP
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3202428 Matrix: Water

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/L	<0.010	0.010	09/20/22 01:26	
Barium	mg/L	<0.50	0.50	09/20/22 01:26	
Cadmium	mg/L	<0.0050	0.0050	09/20/22 01:26	
Chromium	mg/L	<0.010	0.010	09/20/22 01:26	
Lead	mg/L	<0.010	0.010	09/20/22 01:26	
Selenium	mg/L	<0.010	0.010	09/20/22 01:26	
Silver	mg/L	<0.010	0.010	09/20/22 01:26	
Zinc	mg/L	<0.020	0.020	09/20/22 01:26	

LABORATORY CONTROL SAMPLE: 3202429

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	1	1.0	100	80-120	
Barium	mg/L	1	1.0	105	80-120	
Cadmium	mg/L	1	1.0	100	80-120	
Chromium	mg/L	1	0.99	99	80-120	
Lead	mg/L	1	0.98	98	80-120	
Selenium	mg/L	1	1.0	101	80-120	
Silver	mg/L	0.5	0.47	94	80-120	
Zinc	mg/L	1	1.0	101	80-120	

MATRIX SPIKE SAMPLE: 3202430

Parameter	Units	50325322001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.3	103	50-150	
Barium	mg/L	ND	10	10.6	105	50-150	
Cadmium	mg/L	2.5	10	12.7	102	50-150	
Chromium	mg/L	0.10	10	10.2	100	50-150	
Lead	mg/L	ND	10	10.0	100	50-150	
Selenium	mg/L	ND	10	10.3	103	50-150	
Silver	mg/L	ND	5	4.8	96	50-150	
Zinc	mg/L	1600 ug/L	10	12.0	104	50-150	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

MATRIX SPIKE SAMPLE:		3202431					
Parameter	Units	50325351001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	1	1.1	107	50-150	
Barium	mg/L	ND	1	1.1	104	50-150	
Cadmium	mg/L	ND	1	1.0	102	50-150	
Chromium	mg/L	ND	1	0.96	96	50-150	
Lead	mg/L	ND	1	0.91	91	50-150	
Selenium	mg/L	ND	1	1.1	107	50-150	
Silver	mg/L	ND	0.5	0.48	97	50-150	
Zinc	mg/L	27.9 ug/L	1	1.1	103	50-150	

MATRIX SPIKE SAMPLE:		3202432					
Parameter	Units	50325730001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.1	101	50-150	
Barium	mg/L	ND	10	10.6	103	50-150	
Cadmium	mg/L	ND	10	10.0	100	50-150	
Chromium	mg/L	ND	10	9.7	97	50-150	
Lead	mg/L	ND	10	9.5	95	50-150	
Selenium	mg/L	ND	10	10.2	101	50-150	
Silver	mg/L	ND	5	4.7	95	50-150	
Zinc	mg/L	ND	10	9.9	99	50-150	

MATRIX SPIKE SAMPLE:		3202433					
Parameter	Units	50325589001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.2	102	50-150	
Barium	mg/L	ND	10	10.6	104	50-150	
Cadmium	mg/L	ND	10	9.9	99	50-150	
Chromium	mg/L	ND	10	9.6	96	50-150	
Lead	mg/L	4.9	10	14.4	95	50-150	
Selenium	mg/L	ND	10	10.1	101	50-150	
Silver	mg/L	ND	5	4.6	93	50-150	
Zinc	mg/L	127000 ug/L	10	132	48	50-150	P6

MATRIX SPIKE SAMPLE:		3202434					
Parameter	Units	50325450001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.0	100	50-150	
Barium	mg/L	ND	10	10.3	102	50-150	
Cadmium	mg/L	ND	10	9.8	97	50-150	
Chromium	mg/L	ND	10	9.4	94	50-150	
Lead	mg/L	ND	10	9.5	95	50-150	
Selenium	mg/L	ND	10	10.1	100	50-150	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

MATRIX SPIKE SAMPLE:		3202434					
Parameter	Units	50325450001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Silver	mg/L	ND	5	4.6	92	50-150	
Zinc	mg/L	ND	10	10.0	100	50-150	

MATRIX SPIKE SAMPLE:		3202435					
Parameter	Units	50325452001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.2	102	50-150	
Barium	mg/L	ND	10	11.8	104	50-150	
Cadmium	mg/L	ND	10	9.9	99	50-150	
Chromium	mg/L	ND	10	9.5	95	50-150	
Lead	mg/L	ND	10	9.6	96	50-150	
Selenium	mg/L	ND	10	10.2	101	50-150	
Silver	mg/L	ND	5	4.7	93	50-150	
Zinc	mg/L	ND	10	10.2	100	50-150	

MATRIX SPIKE SAMPLE:		3202436					
Parameter	Units	50325420009 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.3	103	50-150	
Barium	mg/L	ND	10	10.7	104	50-150	
Cadmium	mg/L	ND	10	10	100	50-150	
Chromium	mg/L	ND	10	9.6	96	50-150	
Lead	mg/L	ND	10	9.6	96	50-150	
Selenium	mg/L	ND	10	10.2	102	50-150	
Silver	mg/L	ND	5	4.7	95	50-150	
Zinc	mg/L	ND	10	10.1	101	50-150	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE:		3202437			3202438							
Parameter	Units	50325435002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/L	ND	10	10	10.3	10.2	103	102	50-150	1	20	
Barium	mg/L	ND	10	10	11.2	11.2	103	103	50-150	0	20	
Cadmium	mg/L	ND	10	10	10.0	9.9	100	99	50-150	2	20	
Chromium	mg/L	ND	10	10	9.6	9.5	96	95	50-150	1	20	
Lead	mg/L	ND	10	10	9.6	9.5	96	95	50-150	1	20	
Selenium	mg/L	ND	10	10	10.3	10.2	102	101	50-150	1	20	
Silver	mg/L	ND	5	5	4.7	4.7	95	93	50-150	2	20	
Zinc	mg/L	ND	10	10	10.1	10	100	99	50-150	1	20	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

MATRIX SPIKE SAMPLE:		3202439					
Parameter	Units	50325469001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.10	10	10.0	100	50-150	
Barium	mg/L	<5.0	10	10.3	103	50-150	
Cadmium	mg/L	<0.050	10	9.8	98	50-150	
Chromium	mg/L	<0.10	10	9.6	95	50-150	
Lead	mg/L	<0.10	10	9.6	96	50-150	
Selenium	mg/L	<0.10	10	10.1	101	50-150	
Silver	mg/L	<0.10	5	4.6	92	50-150	
Zinc	mg/L	2.3	10	12.4	101	50-150	

MATRIX SPIKE SAMPLE:		3202440					
Parameter	Units	50325535001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.0	100	50-150	
Barium	mg/L	ND	10	10.2	101	50-150	
Cadmium	mg/L	ND	10	9.7	97	50-150	
Chromium	mg/L	ND	10	9.3	93	50-150	
Lead	mg/L	ND	10	9.4	94	50-150	
Selenium	mg/L	ND	10	10	99	50-150	
Silver	mg/L	ND	5	4.5	91	50-150	
Zinc	mg/L	ND	10	9.9	98	50-150	

MATRIX SPIKE SAMPLE:		3202441					
Parameter	Units	50325537001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.0	100	50-150	
Barium	mg/L	ND	10	10.6	103	50-150	
Cadmium	mg/L	ND	10	9.7	97	50-150	
Chromium	mg/L	ND	10	9.4	94	50-150	
Lead	mg/L	ND	10	9.6	95	50-150	
Selenium	mg/L	ND	10	10.1	101	50-150	
Silver	mg/L	ND	5	4.6	92	50-150	
Zinc	mg/L	381 ug/L	10	10.4	100	50-150	

MATRIX SPIKE SAMPLE:		3202442					
Parameter	Units	50325597001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.1	101	50-150	
Barium	mg/L	ND	10	10.5	104	50-150	
Cadmium	mg/L	ND	10	9.9	99	50-150	
Chromium	mg/L	ND	10	9.6	96	50-150	
Lead	mg/L	ND	10	9.7	97	50-150	
Selenium	mg/L	ND	10	10.1	101	50-150	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

MATRIX SPIKE SAMPLE:		3202442					
		50325597001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Silver	mg/L	ND	5	4.7	93	50-150	
Zinc	mg/L	137000 ug/L	10	142	51	50-150	

MATRIX SPIKE SAMPLE:		3202443					
		50325598001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Arsenic	mg/L	ND	10	9.9	99	50-150	
Barium	mg/L	ND	10	10.3	102	50-150	
Cadmium	mg/L	ND	10	9.7	96	50-150	
Chromium	mg/L	ND	10	9.3	93	50-150	
Lead	mg/L	ND	10	9.5	95	50-150	
Selenium	mg/L	ND	10	9.9	99	50-150	
Silver	mg/L	ND	5	4.5	91	50-150	
Zinc	mg/L	1130 ug/L	10	11.1	100	50-150	

MATRIX SPIKE SAMPLE:		3202444					
		50325599001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Arsenic	mg/L	0.16	10	10.3	101	50-150	
Barium	mg/L	ND	10	10.5	104	50-150	
Cadmium	mg/L	ND	10	9.9	99	50-150	
Chromium	mg/L	0.11	10	9.7	96	50-150	
Lead	mg/L	ND	10	9.7	97	50-150	
Selenium	mg/L	ND	10	10.1	101	50-150	
Silver	mg/L	ND	5	4.7	94	50-150	
Zinc	mg/L	547 ug/L	10	10.7	102	50-150	

MATRIX SPIKE SAMPLE:		3202445					
		50325600001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Arsenic	mg/L	ND	10	10	100	50-150	
Barium	mg/L	ND	10	10.5	102	50-150	
Cadmium	mg/L	ND	10	9.7	97	50-150	
Chromium	mg/L	ND	10	9.4	93	50-150	
Lead	mg/L	ND	10	9.5	95	50-150	
Selenium	mg/L	ND	10	10	100	50-150	
Silver	mg/L	ND	5	4.6	91	50-150	
Zinc	mg/L	2610 ug/L	10	12.5	99	50-150	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

MATRIX SPIKE SAMPLE:		3202446					
Parameter	Units	50325601001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.2	102	50-150	
Barium	mg/L	ND	10	10.5	104	50-150	
Cadmium	mg/L	ND	10	9.8	98	50-150	
Chromium	mg/L	ND	10	9.4	94	50-150	
Lead	mg/L	ND	10	9.7	97	50-150	
Selenium	mg/L	ND	10	10.1	101	50-150	
Silver	mg/L	ND	5	4.7	93	50-150	
Zinc	mg/L	99500 ug/L	10	105	51	50-150	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

QC Batch:	695969	Analysis Method:	EPA 6020
QC Batch Method:	EPA 3050B	Analysis Description:	6020 MET
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3199729 Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.10	0.10	09/19/22 15:11	
Cadmium	mg/kg	<0.050	0.050	09/19/22 15:11	
Chromium	mg/kg	<0.20	0.20	09/19/22 15:11	
Copper	mg/kg	<0.10	0.10	09/19/22 15:11	
Lead	mg/kg	<0.10	0.10	09/19/22 15:11	
Molybdenum	mg/kg	<0.10	0.10	09/19/22 15:11	N2
Nickel	mg/kg	<0.050	0.050	09/19/22 15:11	
Selenium	mg/kg	<0.10	0.10	09/19/22 15:11	
Silver	mg/kg	<0.050	0.050	09/19/22 15:11	
Zinc	mg/kg	<0.50	0.50	09/19/22 15:11	

LABORATORY CONTROL SAMPLE: 3199730

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	4	3.9	98	80-120	
Cadmium	mg/kg	4	4.0	99	80-120	
Chromium	mg/kg	4	4.1	103	80-120	
Copper	mg/kg	4	3.9	98	80-120	
Lead	mg/kg	4	4.0	100	80-120	
Molybdenum	mg/kg	4	4.1	102	80-120	N2
Nickel	mg/kg	4	3.9	98	80-120	
Selenium	mg/kg	4	3.9	99	80-120	
Silver	mg/kg	4	4.1	102	80-120	
Zinc	mg/kg	4	3.9	98	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3199731 3199732

Parameter	Units	52103736001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	ND	8	7.9	8.2	8.2	103	103	75-125	1	20	
Cadmium	mg/kg	ND	8	7.9	7.9	7.8	99	98	75-125	2	20	
Chromium	mg/kg	ND	8	7.9	8.2	8.2	102	102	75-125	0	20	
Copper	mg/kg	ND	8	7.9	8.0	8.0	98	98	75-125	0	20	
Lead	mg/kg	ND	8	7.9	7.9	7.9	99	100	75-125	0	20	
Molybdenum	mg/kg	ND	8	7.9	8.1	8.1	102	103	75-125	0	20	N2
Nickel	mg/kg	ND	8	7.9	7.9	7.8	98	98	75-125	1	20	
Selenium	mg/kg	ND	8	7.9	8.9	9.2	111	115	75-125	3	20	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3199731 3199732												
Parameter	Units	52103736001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Silver	mg/kg	ND	8	7.9	8.1	8.0	102	101	75-125	2	20	
Zinc	mg/kg	ND	8	7.9	8.1	8.1	96	96	75-125	1	20	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

QC Batch:	696426	Analysis Method:	EPA 8260
QC Batch Method:	EPA 8260	Analysis Description:	8260 MSV 5030 Low
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3202408 Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,1,1-Trichloroethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,1,2,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,1,2-Trichloroethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,1-Dichloroethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,1-Dichloroethene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,2,3-Trichloropropane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,2-Dibromo-3-chloropropane	mg/kg	<0.010	0.010	09/19/22 11:01	
1,2-Dibromoethane (EDB)	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,2-Dichlorobenzene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,2-Dichloroethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,2-Dichloropropane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,3-Dichlorobenzene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
1,4-Dichlorobenzene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
2-Butanone (MEK)	mg/kg	<0.025	0.025	09/19/22 11:01	
2-Hexanone	mg/kg	<0.10	0.10	09/19/22 11:01	
4-Methyl-2-pentanone (MIBK)	mg/kg	<0.025	0.025	09/19/22 11:01	
Acetone	mg/kg	<0.10	0.10	09/19/22 11:01	
Acrolein	mg/kg	<0.10	0.10	09/19/22 11:01	
Acrylonitrile	mg/kg	<0.10	0.10	09/19/22 11:01	
Benzene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Bromodichloromethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Bromoform	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Bromomethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Carbon disulfide	mg/kg	<0.010	0.010	09/19/22 11:01	
Carbon tetrachloride	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Chlorobenzene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Chloroethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Chloroform	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Chloromethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
cis-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
cis-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Dibromochloromethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Dibromomethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Dichlorodifluoromethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Ethyl methacrylate	mg/kg	<0.10	0.10	09/19/22 11:01	
Ethylbenzene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Iodomethane	mg/kg	<0.10	0.10	09/19/22 11:01	
Methyl methacrylate	mg/kg	<0.10	0.10	09/19/22 11:01	
Methyl-tert-butyl ether	mg/kg	<0.0050	0.0050	09/19/22 11:01	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

METHOD BLANK: 3202408

Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Methylene Chloride	mg/kg	<0.020	0.020	09/19/22 11:01	
Styrene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Tetrachloroethene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Toluene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
trans-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
trans-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
trans-1,4-Dichloro-2-butene	mg/kg	<0.10	0.10	09/19/22 11:01	
Trichloroethene	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Trichlorofluoromethane	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Vinyl acetate	mg/kg	<0.10	0.10	09/19/22 11:01	
Vinyl chloride	mg/kg	<0.0050	0.0050	09/19/22 11:01	
Xylene (Total)	mg/kg	<0.010	0.010	09/19/22 11:01	
4-Bromofluorobenzene (S)	%	93	63-129	09/19/22 11:01	
Dibromofluoromethane (S)	%	111	62-146	09/19/22 11:01	1d
Toluene-d8 (S)	%	94	68-143	09/19/22 11:01	

LABORATORY CONTROL SAMPLE: 3202409

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	0.05	0.051	101	73-121	
1,1,1-Trichloroethane	mg/kg	0.05	0.057	115	60-122	
1,1,2,2-Tetrachloroethane	mg/kg	0.05	0.051	102	60-129	
1,1,2-Trichloroethane	mg/kg	0.05	0.053	106	69-126	
1,1-Dichloroethane	mg/kg	0.05	0.054	108	62-124	
1,1-Dichloroethene	mg/kg	0.05	0.061	123	57-133	
1,2,3-Trichloropropane	mg/kg	0.05	0.058	116	68-120	
1,2-Dibromo-3-chloropropane	mg/kg	0.05	0.053	107	56-133	
1,2-Dibromoethane (EDB)	mg/kg	0.05	0.056	112	62-134	
1,2-Dichlorobenzene	mg/kg	0.05	0.050	100	65-116	
1,2-Dichloroethane	mg/kg	0.05	0.055	111	63-127	
1,2-Dichloropropane	mg/kg	0.05	0.054	108	64-124	
1,3-Dichlorobenzene	mg/kg	0.05	0.050	101	60-118	
1,4-Dichlorobenzene	mg/kg	0.05	0.050	101	59-117	
2-Butanone (MEK)	mg/kg	0.25	0.28	113	53-123	
2-Hexanone	mg/kg	0.25	0.25	101	50-126	
4-Methyl-2-pentanone (MIBK)	mg/kg	0.25	0.25	100	54-130	
Acetone	mg/kg	0.25	0.33	132	21-145	
Acrolein	mg/kg	1	0.77	77	22-129	
Acrylonitrile	mg/kg	0.25	0.29	115	61-125	
Benzene	mg/kg	0.05	0.052	105	65-124	
Bromodichloromethane	mg/kg	0.05	0.057	113	71-121	
Bromoform	mg/kg	0.05	0.054	108	65-120	
Bromomethane	mg/kg	0.05	0.075	150	35-155	
Carbon disulfide	mg/kg	0.05	0.056	112	50-125	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

LABORATORY CONTROL SAMPLE: 3202409

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Carbon tetrachloride	mg/kg	0.05	0.061	121	63-129	
Chlorobenzene	mg/kg	0.05	0.050	100	64-118	
Chloroethane	mg/kg	0.05	0.058	117	41-144	
Chloroform	mg/kg	0.05	0.055	109	60-118	
Chloromethane	mg/kg	0.05	0.051	102	34-127	
cis-1,2-Dichloroethene	mg/kg	0.05	0.057	114	65-121	
cis-1,3-Dichloropropene	mg/kg	0.05	0.051	103	69-124	
Dibromochloromethane	mg/kg	0.05	0.052	104	69-126	
Dibromomethane	mg/kg	0.05	0.058	115	71-120	
Dichlorodifluoromethane	mg/kg	0.05	0.070	140	10-134	L1
Ethyl methacrylate	mg/kg	0.05	<0.10	107	63-123	
Ethylbenzene	mg/kg	0.05	0.049	98	63-119	
Iodomethane	mg/kg	0.05	<0.10	129	37-149	
Methyl methacrylate	mg/kg	0.05	<0.10	120	62-141	
Methyl-tert-butyl ether	mg/kg	0.05	0.059	118	63-128	
Methylene Chloride	mg/kg	0.05	0.058	117	54-141	
Styrene	mg/kg	0.05	0.049	97	65-119	
Tetrachloroethene	mg/kg	0.05	0.053	106	60-122	
Toluene	mg/kg	0.05	0.048	96	61-117	
trans-1,2-Dichloroethene	mg/kg	0.05	0.057	113	61-121	
trans-1,3-Dichloropropene	mg/kg	0.05	0.050	100	68-122	
trans-1,4-Dichloro-2-butene	mg/kg	0.05	<0.10	102	52-140	
Trichloroethene	mg/kg	0.05	0.056	111	63-123	
Trichlorofluoromethane	mg/kg	0.05	0.070	139	44-137	L1
Vinyl acetate	mg/kg	0.2	0.20	101	36-96	L1
Vinyl chloride	mg/kg	0.05	0.062	123	37-136	
Xylene (Total)	mg/kg	0.15	0.15	97	61-120	
4-Bromofluorobenzene (S)	%			92	63-129	
Dibromofluoromethane (S)	%			107	62-146	
Toluene-d8 (S)	%			90	68-143	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

QC Batch: 696419	Analysis Method: EPA 5030/8260
QC Batch Method: EPA 5030/8260	Analysis Description: 8260 MSV TCLP
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3202384 Matrix: Water
Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1-Dichloroethene	mg/L	<0.050	0.050	09/19/22 11:04	
1,2-Dichloroethane	mg/L	<0.050	0.050	09/19/22 11:04	
2-Butanone (MEK)	mg/L	<1.0	1.0	09/19/22 11:04	
Benzene	mg/L	<0.050	0.050	09/19/22 11:04	
Carbon tetrachloride	mg/L	<0.050	0.050	09/19/22 11:04	
Chlorobenzene	mg/L	<0.050	0.050	09/19/22 11:04	
Chloroform	mg/L	<0.050	0.050	09/19/22 11:04	
Tetrachloroethene	mg/L	<0.050	0.050	09/19/22 11:04	
Trichloroethene	mg/L	<0.050	0.050	09/19/22 11:04	
Vinyl chloride	mg/L	<0.020	0.020	09/19/22 11:04	
4-Bromofluorobenzene (S)	%	93	79-124	09/19/22 11:04	
Dibromofluoromethane (S)	%	113	82-128	09/19/22 11:04	
Toluene-d8 (S)	%	100	73-122	09/19/22 11:04	

LABORATORY CONTROL SAMPLE: 3202385

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	0.5	0.59	118	63-138	
1,2-Dichloroethane	mg/L	0.5	0.54	109	68-126	
2-Butanone (MEK)	mg/L	2.5	2.4	97	57-130	
Benzene	mg/L	0.5	0.54	108	76-121	
Carbon tetrachloride	mg/L	0.5	0.62	124	71-134	
Chlorobenzene	mg/L	0.5	0.57	113	74-119	
Chloroform	mg/L	0.5	0.56	111	68-123	
Tetrachloroethene	mg/L	0.5	0.58	116	74-129	
Trichloroethene	mg/L	0.5	0.58	116	73-125	
Vinyl chloride	mg/L	0.5	0.43	86	46-134	
4-Bromofluorobenzene (S)	%			98	79-124	
Dibromofluoromethane (S)	%			107	82-128	
Toluene-d8 (S)	%			104	73-122	

MATRIX SPIKE SAMPLE: 3202386

Parameter	Units	50325435002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.5	0.62	124	55-158	
1,2-Dichloroethane	mg/L	ND	0.5	0.56	113	61-144	
2-Butanone (MEK)	mg/L	ND	2.5	2.4	97	49-149	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

MATRIX SPIKE SAMPLE:		3202386					
Parameter	Units	50325435002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Benzene	mg/L	ND	0.5	0.56	112	68-139	
Carbon tetrachloride	mg/L	ND	0.5	0.69	138	61-153	
Chlorobenzene	mg/L	ND	0.5	0.58	117	57-137	
Chloroform	mg/L	ND	0.5	0.59	117	61-138	
Tetrachloroethene	mg/L	ND	0.5	0.60	120	50-149	
Trichloroethene	mg/L	ND	0.5	0.59	117	55-147	
Vinyl chloride	mg/L	ND	0.5	0.46	92	36-154	
4-Bromofluorobenzene (S)	%				94	79-124	
Dibromofluoromethane (S)	%				105	82-128	
Toluene-d8 (S)	%				103	73-122	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

QC Batch: 695675	Analysis Method: EPA 8081
QC Batch Method: EPA 3546	Analysis Description: 8081 GCS Pesticides
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3198476 Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
4,4'-DDD	ug/kg	<5.0	5.0	09/14/22 22:12	
4,4'-DDE	ug/kg	<5.0	5.0	09/14/22 22:12	
4,4'-DDT	ug/kg	<5.0	5.0	09/14/22 22:12	
Aldrin	ug/kg	<2.5	2.5	09/14/22 22:12	
alpha-BHC	ug/kg	<2.5	2.5	09/14/22 22:12	
alpha-Chlordane	ug/kg	<2.5	2.5	09/14/22 22:12	
beta-BHC	ug/kg	<2.5	2.5	09/14/22 22:12	
Chlordane (Technical)	ug/kg	<50.0	50.0	09/14/22 22:12	
delta-BHC	ug/kg	<2.5	2.5	09/14/22 22:12	
Dieldrin	ug/kg	<5.0	5.0	09/14/22 22:12	
Endosulfan I	ug/kg	<2.5	2.5	09/14/22 22:12	
Endosulfan II	ug/kg	<5.0	5.0	09/14/22 22:12	
Endosulfan sulfate	ug/kg	<5.0	5.0	09/14/22 22:12	
Endrin	ug/kg	<5.0	5.0	09/14/22 22:12	
Endrin aldehyde	ug/kg	<5.0	5.0	09/14/22 22:12	
gamma-BHC (Lindane)	ug/kg	<2.5	2.5	09/14/22 22:12	
gamma-Chlordane	ug/kg	<2.5	2.5	09/14/22 22:12	
Heptachlor	ug/kg	<2.5	2.5	09/14/22 22:12	
Heptachlor epoxide	ug/kg	<2.5	2.5	09/14/22 22:12	
Toxaphene	ug/kg	<50.0	50.0	09/14/22 22:12	
Decachlorobiphenyl (S)	%	92	10-147	09/14/22 22:12	

LABORATORY CONTROL SAMPLE: 3198477

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
4,4'-DDD	ug/kg	19.7	19.0	97	40-148	
4,4'-DDE	ug/kg	19.7	17.1	87	42-145	
4,4'-DDT	ug/kg	19.7	17.2	87	28-169	
Aldrin	ug/kg	9.9	8.8	89	42-137	
alpha-BHC	ug/kg	9.9	8.7	89	40-141	
alpha-Chlordane	ug/kg	9.9	9.7	99	38-141	
beta-BHC	ug/kg	9.9	9.3	95	42-137	
delta-BHC	ug/kg	9.9	5.4	55	33-132	
Dieldrin	ug/kg	19.7	19.3	98	42-143	
Endosulfan I	ug/kg	9.9	8.9	90	38-142	
Endosulfan II	ug/kg	19.7	18.4	93	42-153	
Endosulfan sulfate	ug/kg	19.7	17.2	87	36-133	
Endrin	ug/kg	19.7	19.0	96	43-145	
Endrin aldehyde	ug/kg	19.7	22.3	113	40-138	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

LABORATORY CONTROL SAMPLE: 3198477

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-BHC (Lindane)	ug/kg	9.9	9.0	91	42-142	
gamma-Chlordane	ug/kg	9.9	10.0	102	44-148	
Heptachlor	ug/kg	9.9	9.4	95	42-143	
Heptachlor epoxide	ug/kg	9.9	9.3	94	41-136	
Decachlorobiphenyl (S)	%			105	10-147	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3198478 3198479

Parameter	Units	60409974001		MS		MSD		MS		MSD		% Rec		Max		Qual
		Result	Conc.	Spike Conc.	Conc.	Result	Conc.	Result	Conc.	Result	Conc.	Limits	RPD	RPD	RPD	
4,4'-DDD	ug/kg	<1.1	21.5	22.1	22.1	16.5	15.8	76	71	10-171	5	20				
4,4'-DDE	ug/kg	<0.92	21.5	22.1	22.1	13.2	12.7	61	57	10-146	4	20				
4,4'-DDT	ug/kg	<1.8	21.5	22.1	22.1	12.5	10.0	58	45	10-165	22	20	R1			
Aldrin	ug/kg	<0.54	10.8	11.1	11.1	7.3	7.1	67	64	12-141	3	20				
alpha-BHC	ug/kg	<0.58	10.8	11.1	11.1	7.7	7.4	71	67	14-157	3	20				
alpha-Chlordane	ug/kg	<0.49	10.8	11.1	11.1	7.9	7.7	73	69	10-153	3	20				
beta-BHC	ug/kg	<0.52	10.8	11.1	11.1	8.3	8.0	77	73	10-163	4	20				
delta-BHC	ug/kg	<0.73	10.8	11.1	11.1	4.7	4.9	43	44	10-155	4	20				
Dieldrin	ug/kg	<0.98	21.5	22.1	22.1	14.2	13.6	66	61	14-141	5	20				
Endosulfan I	ug/kg	<0.46	10.8	11.1	11.1	6.9	6.5	64	59	10-141	5	20				
Endosulfan II	ug/kg	<0.91	21.5	22.1	22.1	14.1	13.9	65	63	10-149	2	20				
Endosulfan sulfate	ug/kg	<1.1	21.5	22.1	22.1	12.9	12.7	60	57	10-126	2	20				
Endrin	ug/kg	<0.98	21.5	22.1	22.1	14.5	13.8	67	63	13-136	5	20				
Endrin aldehyde	ug/kg	<1.3	21.5	22.1	22.1	14.7	14.6	68	66	10-150	1	20				
gamma-BHC (Lindane)	ug/kg	<0.51	10.8	11.1	11.1	9.3	8.9	86	81	18-152	4	20				
gamma-Chlordane	ug/kg	<0.58	10.8	11.1	11.1	8.0	7.5	74	68	10-159	5	20				
Heptachlor	ug/kg	<0.50	10.8	11.1	11.1	8.5	8.0	79	72	10-162	6	20				
Heptachlor epoxide	ug/kg	<0.47	10.8	11.1	11.1	7.4	7.0	69	64	15-142	5	20				
Decachlorobiphenyl (S)	%							79	61	10-147						

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

QC Batch: 695706	Analysis Method: EPA 8082
QC Batch Method: EPA 3546	Analysis Description: 8082 PCB Solids
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3198571 Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	<96.2	96.2	09/15/22 13:00	
PCB-1221 (Aroclor 1221)	ug/kg	<96.2	96.2	09/15/22 13:00	
PCB-1232 (Aroclor 1232)	ug/kg	<96.2	96.2	09/15/22 13:00	
PCB-1242 (Aroclor 1242)	ug/kg	<96.2	96.2	09/15/22 13:00	
PCB-1248 (Aroclor 1248)	ug/kg	<96.2	96.2	09/15/22 13:00	
PCB-1254 (Aroclor 1254)	ug/kg	<96.2	96.2	09/15/22 13:00	
PCB-1260 (Aroclor 1260)	ug/kg	<96.2	96.2	09/15/22 13:00	
Tetrachloro-m-xylene (S)	%	71	36-112	09/15/22 13:00	

LABORATORY CONTROL SAMPLE: 3198572

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	316	287	91	52-128	
PCB-1260 (Aroclor 1260)	ug/kg	316	232	73	30-128	
Tetrachloro-m-xylene (S)	%			71	36-112	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3198573 3198574

Parameter	Units	50325495001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
PCB-1016 (Aroclor 1016)	ug/kg	ND	1520	1580	<457	<474	11	11	10-150		20	
PCB-1260 (Aroclor 1260)	ug/kg	ND	1520	1580	<457	<474	18	19	10-140		20	
Tetrachloro-m-xylene (S)	%						2	2	36-112		S0	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

QC Batch: 696595

QC Batch Method: EPA 8151

Analysis Method: EPA 8151A

Analysis Description: 8151 GCS TCLP Herbicides

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3202953

Matrix: Water

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0010	0.0010	09/21/22 17:10	
2,4-D	mg/L	<0.0010	0.0010	09/21/22 17:10	
2,4-DCAA (S)	%.	73	12-143	09/21/22 17:10	

LABORATORY CONTROL SAMPLE: 3202954

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	0.005	0.0048	97	27-127	
2,4-D	mg/L	0.005	0.0046	92	10-160	
2,4-DCAA (S)	%.			83	12-143	

MATRIX SPIKE SAMPLE: 3202955

Parameter	Units	50325789001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	ND	0.025	0.015	60	22-150	
2,4-D	mg/L	ND	0.025	0.017	67	11-160	
2,4-DCAA (S)	%.				71	12-143	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

QC Batch:	696599	Analysis Method:	EPA 8270
QC Batch Method:	EPA 3546	Analysis Description:	8270 Solid MSSV Microwave Short Spike
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3202962 Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1-Methylnaphthalene	ug/kg	<328	328	09/20/22 12:47	
2,4,5-Trichlorophenol	ug/kg	<328	328	09/20/22 12:47	
2,4,6-Trichlorophenol	ug/kg	<328	328	09/20/22 12:47	
2,4-Dichlorophenol	ug/kg	<328	328	09/20/22 12:47	
2,4-Dimethylphenol	ug/kg	<328	328	09/20/22 12:47	
2,4-Dinitrophenol	ug/kg	<1590	1590	09/20/22 12:47	
2,4-Dinitrotoluene	ug/kg	<328	328	09/20/22 12:47	
2,6-Dinitrotoluene	ug/kg	<328	328	09/20/22 12:47	
2-Chloronaphthalene	ug/kg	<328	328	09/20/22 12:47	
2-Chlorophenol	ug/kg	<328	328	09/20/22 12:47	
2-Methylnaphthalene	ug/kg	<328	328	09/20/22 12:47	
2-Methylphenol(o-Cresol)	ug/kg	<328	328	09/20/22 12:47	
2-Nitroaniline	ug/kg	<328	328	09/20/22 12:47	
2-Nitrophenol	ug/kg	<328	328	09/20/22 12:47	
3&4-Methylphenol(m&p Cresol)	ug/kg	<656	656	09/20/22 12:47	
3,3'-Dichlorobenzidine	ug/kg	<656	656	09/20/22 12:47	
3-Nitroaniline	ug/kg	<328	328	09/20/22 12:47	
4,6-Dinitro-2-methylphenol	ug/kg	<656	656	09/20/22 12:47	
4-Bromophenylphenyl ether	ug/kg	<328	328	09/20/22 12:47	
4-Chloro-3-methylphenol	ug/kg	<656	656	09/20/22 12:47	
4-Chloroaniline	ug/kg	<656	656	09/20/22 12:47	
4-Chlorophenylphenyl ether	ug/kg	<328	328	09/20/22 12:47	
4-Nitroaniline	ug/kg	<328	328	09/20/22 12:47	
4-Nitrophenol	ug/kg	<1590	1590	09/20/22 12:47	
Acenaphthene	ug/kg	<328	328	09/20/22 12:47	
Acenaphthylene	ug/kg	<328	328	09/20/22 12:47	
Anthracene	ug/kg	<328	328	09/20/22 12:47	
Benzo(a)anthracene	ug/kg	<328	328	09/20/22 12:47	
Benzo(a)pyrene	ug/kg	<328	328	09/20/22 12:47	
Benzo(b)fluoranthene	ug/kg	<328	328	09/20/22 12:47	
Benzo(g,h,i)perylene	ug/kg	<328	328	09/20/22 12:47	
Benzo(k)fluoranthene	ug/kg	<328	328	09/20/22 12:47	
Benzyl alcohol	ug/kg	<656	656	09/20/22 12:47	
bis(2-Chloroethoxy)methane	ug/kg	<328	328	09/20/22 12:47	
bis(2-Chloroethyl) ether	ug/kg	<328	328	09/20/22 12:47	
bis(2-Ethylhexyl)phthalate	ug/kg	<328	328	09/20/22 12:47	
bis(2chloro1methylethyl) ether	ug/kg	<328	328	09/20/22 12:47	
Butylbenzylphthalate	ug/kg	<328	328	09/20/22 12:47	
Chrysene	ug/kg	<328	328	09/20/22 12:47	
Di-n-butylphthalate	ug/kg	<328	328	09/20/22 12:47	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

METHOD BLANK: 3202962

Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Di-n-octylphthalate	ug/kg	<328	328	09/20/22 12:47	
Dibenz(a,h)anthracene	ug/kg	<328	328	09/20/22 12:47	
Dibenzofuran	ug/kg	<328	328	09/20/22 12:47	
Diethylphthalate	ug/kg	<328	328	09/20/22 12:47	
Dimethylphthalate	ug/kg	<328	328	09/20/22 12:47	
Fluoranthene	ug/kg	<328	328	09/20/22 12:47	
Fluorene	ug/kg	<328	328	09/20/22 12:47	
Hexachloro-1,3-butadiene	ug/kg	<328	328	09/20/22 12:47	
Hexachlorobenzene	ug/kg	<328	328	09/20/22 12:47	
Hexachlorocyclopentadiene	ug/kg	<328	328	09/20/22 12:47	
Hexachloroethane	ug/kg	<328	328	09/20/22 12:47	
Indeno(1,2,3-cd)pyrene	ug/kg	<328	328	09/20/22 12:47	
Isophorone	ug/kg	<328	328	09/20/22 12:47	
N-Nitroso-di-n-propylamine	ug/kg	<328	328	09/20/22 12:47	
N-Nitrosodiphenylamine	ug/kg	<328	328	09/20/22 12:47	
Naphthalene	ug/kg	<328	328	09/20/22 12:47	
Nitrobenzene	ug/kg	<328	328	09/20/22 12:47	
Pentachlorophenol	ug/kg	<1590	1590	09/20/22 12:47	
Phenanthrene	ug/kg	<328	328	09/20/22 12:47	
Phenol	ug/kg	<328	328	09/20/22 12:47	
Pyrene	ug/kg	<328	328	09/20/22 12:47	
2,4,6-Tribromophenol (S)	%	97	10-123	09/20/22 12:47	
2-Fluorobiphenyl (S)	%	79	36-100	09/20/22 12:47	
2-Fluorophenol (S)	%	95	22-114	09/20/22 12:47	
Nitrobenzene-d5 (S)	%	85	35-110	09/20/22 12:47	
p-Terphenyl-d14 (S)	%	95	29-117	09/20/22 12:47	
Phenol-d5 (S)	%	97	35-115	09/20/22 12:47	

LABORATORY CONTROL SAMPLE: 3202963

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4-Dinitrotoluene	ug/kg	1660	1740	105	57-115	
2-Chlorophenol	ug/kg	1660	1500	90	55-103	
2-Methylnaphthalene	ug/kg	1660	1600	97	53-108	
4-Chloro-3-methylphenol	ug/kg	1660	1970	119	60-121	
4-Nitrophenol	ug/kg	1660	1830	110	48-129	
Acenaphthene	ug/kg	1660	1410	85	57-102	
Acenaphthylene	ug/kg	1660	1350	82	56-103	
Anthracene	ug/kg	1660	1490	90	62-106	
Benzo(a)anthracene	ug/kg	1660	1530	93	63-110	
Benzo(a)pyrene	ug/kg	1660	1390	84	60-114	
Benzo(b)fluoranthene	ug/kg	1660	1470	89	61-119	
Benzo(g,h,i)perylene	ug/kg	1660	1560	94	62-109	
Benzo(k)fluoranthene	ug/kg	1660	1560	94	59-115	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

LABORATORY CONTROL SAMPLE: 3202963

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chrysene	ug/kg	1660	1580	95	61-109	
Dibenz(a,h)anthracene	ug/kg	1660	1580	95	62-111	
Fluoranthene	ug/kg	1660	1600	97	65-113	
Fluorene	ug/kg	1660	1530	92	60-109	
Indeno(1,2,3-cd)pyrene	ug/kg	1660	1690	102	62-111	
N-Nitroso-di-n-propylamine	ug/kg	1660	1520	92	51-105	
Naphthalene	ug/kg	1660	1470	89	53-103	
Pentachlorophenol	ug/kg	1660	<1590	84	33-128	
Phenanthrene	ug/kg	1660	1500	90	62-108	
Phenol	ug/kg	1660	1700	103	45-112	
Pyrene	ug/kg	1660	1600	97	61-113	
2,4,6-Tribromophenol (S)	%			96	10-123	
2-Fluorobiphenyl (S)	%			76	36-100	
2-Fluorophenol (S)	%			94	22-114	
Nitrobenzene-d5 (S)	%			81	35-110	
p-Terphenyl-d14 (S)	%			93	29-117	
Phenol-d5 (S)	%			99	35-115	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3202964 3202965

Parameter	Units	50325723007 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
2,4-Dinitrotoluene	ug/kg	ND	2040	2050	1650	748	81	37	10-126	75	20	R1
2-Chlorophenol	ug/kg	ND	2040	2050	1610	989	79	48	15-119	48	20	R1
2-Methylnaphthalene	ug/kg	ND	2040	2050	1710	1260	84	62	24-122	31	20	R1
4-Chloro-3-methylphenol	ug/kg	ND	2040	2050	1920	882	95	43	20-139	74	20	R1
4-Nitrophenol	ug/kg	ND	2040	2050	<1950	<1960	85	37	10-143		20	
Acenaphthene	ug/kg	ND	2040	2050	1460	1030	72	50	18-123	34	20	R1
Acenaphthylene	ug/kg	ND	2040	2050	1390	893	68	44	17-120	43	20	R1
Anthracene	ug/kg	ND	2040	2050	1410	823	69	40	16-126	53	20	R1
Benzo(a)anthracene	ug/kg	ND	2040	2050	1390	589	68	29	10-145	81	20	R1
Benzo(a)pyrene	ug/kg	ND	2040	2050	1160	459	57	22	10-136	86	20	R1
Benzo(b)fluoranthene	ug/kg	ND	2040	2050	1330	496	66	24	10-147	92	20	R1
Benzo(g,h,i)perylene	ug/kg	ND	2040	2050	1280	491	63	24	10-128	89	20	R1
Benzo(k)fluoranthene	ug/kg	ND	2040	2050	1270	528	62	26	10-137	82	20	R1
Chrysene	ug/kg	ND	2040	2050	1420	626	70	31	10-141	77	20	R1
Dibenz(a,h)anthracene	ug/kg	ND	2040	2050	1300	488	64	24	10-130	91	20	R1
Fluoranthene	ug/kg	ND	2040	2050	1420	717	70	35	10-151	66	20	R1
Fluorene	ug/kg	ND	2040	2050	1520	1010	75	49	17-131	40	20	R1
Indeno(1,2,3-cd)pyrene	ug/kg	ND	2040	2050	1380	526	68	26	10-132	90	20	R1
N-Nitroso-di-n-propylamine	ug/kg	ND	2040	2050	1610	1030	79	50	21-118	44	20	R1
Naphthalene	ug/kg	ND	2040	2050	1580	1180	78	58	21-122	29	20	R1
Pentachlorophenol	ug/kg	ND	2040	2050	<1950	<1960	60	26	10-125		20	
Phenanthrene	ug/kg	ND	2040	2050	1460	890	72	44	10-141	49	20	R1
Phenol	ug/kg	ND	2040	2050	1740	843	86	41	17-120	70	20	R1

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3202964 3202965												
Parameter	Units	50325723007		MS	MSD	MS		MS	MSD	% Rec Limits	Max	
		Result	Conc.	Spike Conc.	Spike Conc.	Result	Result	% Rec	% Rec		RPD	RPD Qual
Pyrene	ug/kg	ND	2040	2050	1490	749	73	37	37	10-150	66	20 R1
2,4,6-Tribromophenol (S)	%.						72	31	31	10-123		
2-Fluorobiphenyl (S)	%.						65	47	47	36-100		
2-Fluorophenol (S)	%.						79	46	46	22-114		
Nitrobenzene-d5 (S)	%.						70	48	48	35-110		
p-Terphenyl-d14 (S)	%.						67	30	30	29-117		
Phenol-d5 (S)	%.						81	40	40	35-115		

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50325469

QC Batch:	696683	Analysis Method:	EPA 8270
QC Batch Method:	EPA 3510	Analysis Description:	8270 TCLP MSSV
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3203265 Matrix: Water

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,4-Dichlorobenzene	mg/L	<0.010	0.010	09/21/22 21:48	
2,4,5-Trichlorophenol	mg/L	<0.050	0.050	09/21/22 21:48	
2,4,6-Trichlorophenol	mg/L	<0.010	0.010	09/21/22 21:48	
2,4-Dinitrotoluene	mg/L	<0.010	0.010	09/21/22 21:48	
2-Methylphenol(o-Cresol)	mg/L	<0.010	0.010	09/21/22 21:48	
3&4-Methylphenol(m&p Cresol)	mg/L	<0.020	0.020	09/21/22 21:48	
Hexachloro-1,3-butadiene	mg/L	<0.010	0.010	09/21/22 21:48	
Hexachlorobenzene	mg/L	<0.010	0.010	09/21/22 21:48	
Hexachloroethane	mg/L	<0.010	0.010	09/21/22 21:48	
Nitrobenzene	mg/L	<0.010	0.010	09/21/22 21:48	
Pentachlorophenol	mg/L	<0.050	0.050	09/21/22 21:48	
Pyridine	mg/L	<0.010	0.010	09/21/22 21:48	
2,4,6-Tribromophenol (S)	%	82	48-137	09/21/22 21:48	
2-Fluorobiphenyl (S)	%	51	38-101	09/21/22 21:48	
2-Fluorophenol (S)	%	42	24-70	09/21/22 21:48	
Nitrobenzene-d5 (S)	%	63	51-109	09/21/22 21:48	
p-Terphenyl-d14 (S)	%	89	65-159	09/21/22 21:48	
Phenol-d5 (S)	%	27	16-44	09/21/22 21:48	

LABORATORY CONTROL SAMPLE: 3203266

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	0.05	0.020	39	36-77	
2,4,5-Trichlorophenol	mg/L	0.05	<0.050	83	48-118	
2,4,6-Trichlorophenol	mg/L	0.05	0.040	80	43-122	
2,4-Dinitrotoluene	mg/L	0.05	0.039	79	46-120	
2-Methylphenol(o-Cresol)	mg/L	0.05	0.029	58	45-91	
3&4-Methylphenol(m&p Cresol)	mg/L	0.1	0.054	54	39-88	
Hexachloro-1,3-butadiene	mg/L	0.05	0.020	39	22-81	
Hexachlorobenzene	mg/L	0.05	0.027	53	43-97	
Hexachloroethane	mg/L	0.05	0.018	37	28-75	
Nitrobenzene	mg/L	0.05	0.040	79	55-104	
Pentachlorophenol	mg/L	0.05	<0.050	82	37-116	
Pyridine	mg/L	0.05	0.016	32	11-78	
2,4,6-Tribromophenol (S)	%			90	48-137	
2-Fluorobiphenyl (S)	%			55	38-101	
2-Fluorophenol (S)	%			40	24-70	
Nitrobenzene-d5 (S)	%			76	51-109	
p-Terphenyl-d14 (S)	%			91	65-159	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

LABORATORY CONTROL SAMPLE: 3203266

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phenol-d5 (S)	%.			24	16-44	

MATRIX SPIKE SAMPLE: 3203267

Parameter	Units	50325435001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	ND	0.5	0.18	37	31-78	
2,4,5-Trichlorophenol	mg/L	ND	0.5	<0.50	73	44-123	
2,4,6-Trichlorophenol	mg/L	ND	0.5	0.34	68	43-120	
2,4-Dinitrotoluene	mg/L	ND	0.5	0.37	74	46-113	
2-Methylphenol(o-Cresol)	mg/L	ND	0.5	0.27	54	36-98	
3&4-Methylphenol(m&p Cresol)	mg/L	ND	1	0.52	52	25-100	
Hexachloro-1,3-butadiene	mg/L	ND	0.5	0.18	35	14-90	
Hexachlorobenzene	mg/L	ND	0.5	0.25	51	41-93	
Hexachloroethane	mg/L	ND	0.5	0.16	32	11-92	
Nitrobenzene	mg/L	ND	0.5	0.34	67	47-107	
Pentachlorophenol	mg/L	ND	0.5	<0.50	81	30-127	
Pyridine	mg/L	ND	0.5	0.16	31	15-82	
2,4,6-Tribromophenol (S)	%.				86	48-137	
2-Fluorobiphenyl (S)	%.				53	38-101	
2-Fluorophenol (S)	%.				40	24-70	
Nitrobenzene-d5 (S)	%.				64	51-109	
p-Terphenyl-d14 (S)	%.				88	65-159	
Phenol-d5 (S)	%.				26	16-44	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

QC Batch: 695875

QC Batch Method: SM 2540G

Analysis Method: SM 2540G

Analysis Description: Dry Weight/Percent Moisture

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

SAMPLE DUPLICATE: 3199244

Parameter	Units	50325469001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	98.9	99.0	0	5	N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

QC Batch: 695873

QC Batch Method: SM 2540G

Analysis Method: SM 2540G

Analysis Description: 2540G Total Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

SAMPLE DUPLICATE: 3199241

Parameter	Units	50325469001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	1.1	1.0	2	10	N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

QC Batch: 695872

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Volatile Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

SAMPLE DUPLICATE: 3199239

Parameter	Units	50325469001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	54.7	53.5	2	10	N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

QC Batch: 695435

QC Batch Method: EPA 9038

Analysis Method: EPA 9038

Analysis Description: 9038 Sulfate Soil

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3197465

Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Sulfate	mg/kg	<99.9	99.9	09/13/22 13:05	N2

LABORATORY CONTROL SAMPLE: 3197466

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Sulfate	mg/kg	200	198	99	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3197467 3197468

Parameter	Units	50325273001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Sulfate	mg/kg	116	200	200	358	347	121	116	90-110	3	20	M3, N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

QC Batch: 695712

QC Batch Method: EPA 9045

Analysis Method: EPA 9045

Analysis Description: 9045 pH

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

SAMPLE DUPLICATE: 3198593

Parameter	Units	50325671001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	9.5	9.5	0	2	H3

SAMPLE DUPLICATE: 3198594

Parameter	Units	50325469001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	8.4	8.4	0	2	H3

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

QC Batch: 695211

Analysis Method: EPA 351.2

QC Batch Method: EPA 351.2

Analysis Description: 351.2 TKN

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3196780

Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<48.8	48.8	09/13/22 12:11	N2

LABORATORY CONTROL SAMPLE: 3196781

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	490	493	101	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3196782 3196783

Parameter	Units	50325469001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	159000	46400	46400	222000	227000	135	147	90-110	2	20	M3, N2

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

QC Batch: 695636

Analysis Method: EPA 353.2

QC Batch Method: EPA 353.2

Analysis Description: 353.2 Nitrate + Nitrite

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3198388

Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<5.0	5.0	09/14/22 10:03	N2
Nitrogen, NO2 plus NO3	mg/kg	<5.0	5.0	09/14/22 10:03	N2

LABORATORY CONTROL SAMPLE: 3198389

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	10	10.1	101	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	20	20.6	103	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3198392 3198393

Parameter	Units	50325611001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	ND	3780	3780	3660	3650	97	96	90-110	0	20	N2
Nitrogen, NO2 plus NO3	mg/kg	ND	7560	7560	7450	7440	98	98	90-110	0	20	N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

QC Batch: 695637

Analysis Method: SM 4500-Cl-E

QC Batch Method: SM 4500-Cl-E

Analysis Description: 4500 Chloride

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3198394

Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<99.8	99.8	09/16/22 10:51	N2

LABORATORY CONTROL SAMPLE: 3198395

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	200	202	101	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3198396 3198397

Parameter	Units	50325469001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	<186000	372000	372000	371000	367000	97	96	90-110	1	20	N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50325469

QC Batch: 695422

Analysis Method: SM 4500-CN-E

QC Batch Method: SM 4500-CN-C

Analysis Description: 4500CNE Cyanide

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50325469001

METHOD BLANK: 3197428

Matrix: Solid

Associated Lab Samples: 50325469001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Cyanide	mg/kg	<0.48	0.48	09/13/22 13:42	N2

LABORATORY CONTROL SAMPLE: 3197429

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Cyanide	mg/kg	4.8	4.9	103	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3197430 3197431

Parameter	Units	50324766001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Cyanide	mg/kg	ND	31.6	31.6	33.2	33.4	98	98	90-110	1	20	N2

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QUALIFIERS

Project: Fall Biosolids
Pace Project No.: 50325469

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

ANALYTE QUALIFIERS

1d	A matrix spike/matrix spike duplicate was not performed for this batch due to insufficient sample volume.
CH	The continuing calibration for this compound is outside of Pace Analytical acceptance limits. The results may be biased high.
D3	Sample was diluted due to the presence of high levels of non-target analytes or other matrix interference.
H3	Sample was received or analysis requested beyond the recognized method holding time.
H7	Re-extraction or re-analysis could not be performed within method holding time.
L1	Analyte recovery in the laboratory control sample (LCS) was above QC limits. Results for this analyte in associated samples may be biased high.
M0	Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M3	Matrix spike recovery was outside laboratory control limits due to matrix interferences.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P1	Routine initial sample volume or weight was not used for extraction, resulting in elevated reporting limits.
P6	Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.
R1	RPD value was outside control limits.
S0	Surrogate recovery outside laboratory control limits.
S2	Surrogate recovery outside laboratory control limits due to matrix interferences (confirmed by similar results from sample re-analysis).

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Fall Biosolids

Pace Project No.: 50325469

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50325469001	Digester #4	EPA 3546	695675	EPA 8081	695834
50325469001	Digester #4	EPA 3546	695706	EPA 8082	695835
50325469001	Digester #4	EPA 8151	696595	EPA 8151A	696961
50325469001	Digester #4	EPA 3050	695590	EPA 6010	696214
50325469001	Digester #4	EPA 3010	696431	EPA 6010	696565
50325469001	Digester #4	EPA 3050B	695969	EPA 6020	696227
50325469001	Digester #4	EPA 7470	696493	EPA 7470	696832
50325469001	Digester #4	EPA 7471	696428	EPA 7471	696584
50325469001	Digester #4	EPA 3546	696599	EPA 8270	696706
50325469001	Digester #4	EPA 3510	696683	EPA 8270	697021
50325469001	Digester #4	EPA 8260	696426		
50325469001	Digester #4	EPA 5030/8260	696419		
50325469001	Digester #4	SM 2540G	695875		
50325469001	Digester #4	SM 2540G	695873		
50325469001	Digester #4	SM 2540G	695872		
50325469001	Digester #4	EPA 9038	695435	EPA 9038	695445
50325469001	Digester #4	EPA 9045	695712		
50325469001	Digester #4	EPA 351.2	695211	EPA 351.2	695461
50325469001	Digester #4	EPA 353.2	695636	EPA 353.2	695641
50325469001	Digester #4	SM 4500-CI-E	695637	SM 4500-CI-E	695642
50325469001	Digester #4	SM 4500-CN-C	695422	SM 4500-CN-E	695503

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Sample Conditions Upon Receipt Form (SCUR)

Date/Time: <u>9.9.22</u>		Evaluated By: <u>WDC</u>		WO#: 50325469 PM: BJH Due Date: 09/23/22 CLIENT: GR-Cadillac	
Client: <u>City of Cadillac</u>		PM: <u>BJH</u>			
Lab Notified of Rush or Short Holds: YES ☒ NO ☒					
Project Received Via: FedEx ☒ UPS ☒ Client ☐ Pace Courier ☐ Other: _____					Comments:
Custody Seal Present and Intact:		YES	NO	N/A	
Received Sample Information Form (SIF): Drinking Waters Only		YES	NO	N/A	
Short Hold Present (≤ 48 Hours):		YES	NO		
Sample Received in Hold:		YES	NO		
Custody Signature Present:		YES	NO		
Collector Signature Present:		YES	NO		
Sample Collected Today and On Ice:		YES	NO	N/A	
IR Gun #: 280 <u>281</u>		Temp. should be 0°C - 6°C (Initial/Corrected)			
Ice Type: WET Bagged / WET Loose ☒ BLUE ☐ NONE ☐		1. Cooler Temp. Upon Receipt: <u>1.7/1.3</u> °C			
Ice Location: TOP ☒ BOTTOM ☐ MIDDLE ☐ DISPERSED ☐		2. Cooler Temp. Upon Receipt: _____ °C			
Temp Blank Received:		YES	NO		
Sample Label Matches COC (ID/Date/Time):		YES	NO		
Container Intact:		YES	NO		
Correct Container:		YES	NO		
Sufficient Volume:		YES	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation pH Strip Lot #: _____ <i>Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl</i>		YES	NO	N/A	
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ <i>Applies to SVOC 625, PCB/Pest. 608, Total/Amenable Cyanide</i>		YES	NO	N/A	
VOA Headspace Acceptable (<6mm):		YES	NO	N/A	
Trip Blank Received: HCl ☐ MeOH ☐ Other: _____		YES	NO	ON HOLD	
Comments:		3. Cooler Temp. Upon Receipt: _____ °C			
		4. Cooler Temp. Upon Receipt: _____ °C			
		Non-Conformance Form Required: YES ☒ NO ☐			